

5.1.4 Ideal gases

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) amount of substance in moles; Avogadro constant N_A equals $6.02 \times 10^{23} \text{ mol}^{-1}$	
(b) model of kinetic theory of gases; assumptions – particles (atoms or molecules) occupy negligible volume, that all collisions are perfectly elastic, and that there are negligible forces between particles except during collision	
(c) pressure in terms of this model	HSW1 Explanation of pressure in terms of Newtonian theory.
(d) (i) the equation of state of an ideal gas $pV = nRT$, where n is the number of moles	

- (d) (ii) techniques and procedures used to investigate $PV = \text{constant}$ (Boyle's law) and

$$\frac{P}{T} = \text{constant}$$

- (iii) an estimation of absolute zero using variation of gas temperature with pressure

- (e) the equation $pV = \frac{1}{3}Nm\overline{c^2}$, where N is the number of particles (atoms or molecules) and $\overline{c^2}$ is the mean square speed

- (f) root mean square (r.m.s.) speed; mean square speed

- (g) the Boltzmann constant; $k = \frac{R}{N_A}$

- (h) $pV = NkT$; $\frac{1}{2}m\overline{c^2} = \frac{3}{2}kT$

- (i) internal energy of an ideal gas.

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Derivation of this equation is not required.

Learners should know about the general characteristics of the Maxwell-Boltzmann distribution.

Learners will also be expected to know the derivation of the equation $\frac{1}{2}m\overline{c^2} = \frac{3}{2}kT$ from $pV = \frac{1}{3}Nm\overline{c^2}$ and $pV = NkT$.